代入数据,可知折射线能进入光纤的充要条件是

$$|\beta| \leqslant 33.75^\circ$$

下面分别讨论三条光线的情况.

(i)$\alpha_1 = 30^\circ$ 时

a.此入射光线的折射线能否进入光纤?

由(2)式得 $\sin\theta_1 = 2\sin\alpha_1 = 1$,得

$$\theta_1 = 90^\circ$$

代入(3)式得

$$\theta_1' = \arcsin\left(\frac{2}{3}\right) = 41.81^\circ$$

代入(4)式得

$$\beta_1 = 2\theta_1' - \theta_1 + \alpha_1 = 23.62^\circ \tag{7}$$

因为

$$\beta_1 < \angle QOB \tag{8}$$

故此折射线能进入光纤.

b.此折射线能否在光芯包层的界面上发生全反射?

设光纤内光线射至光芯与包层界面时发生全反射的临界角为 i_C,则

$$i_C = \arcsin\left(\frac{1.400}{1.500}\right) = 68.96^\circ \tag{9}$$

因为 $i = 90^\circ - \alpha'$,所以当 α' 满足下面条件时折射线即能被全反射:

$$\alpha' \leqslant 90^\circ - 68.96^\circ = 21.04^\circ \tag{10}$$

现在 $\alpha_1' = \theta_1 - \alpha_1 - \theta_1' = 18.19^\circ$,满足(10)式的要求.故该光线能经过全反射在光纤中传播.

(ii)$\alpha_2 = 25^\circ$ 时

由(2)式得 $\sin\theta_2 = 2\sin\alpha_2 = 2\sin25^\circ$ $\theta_2 = 57.70^\circ$

$$\sin\theta_2' = \frac{1}{1.5}\sin\theta_2 \qquad \theta_2' = 34.30^\circ$$

$$\beta_2 = 2\theta_2' - \theta_2 + \alpha_2 = 35.90^\circ \tag{11}$$

由(6)式可知,该折射线不能进入 QQ' 右面的光纤.

(iii)$\alpha_3 = 20^\circ$ 时

由(2)式得 $\sin\theta_3 = 2\sin\alpha_3 = 2\sin20^\circ$ $\theta_3 = 43.16^\circ$

$$\sin\theta_3' = \frac{1}{1.5}\sin\theta_3 \qquad \theta_3' = 27.13^\circ$$

$$\beta_3 = 2\theta_3' - \theta_3 + \alpha_3 = 31.10^\circ \tag{12}$$

由(6)式可知,此折射线能进入 QQ' 右面的光纤.

此时

$$\alpha_3' = \theta_3 - \alpha_3 + \theta_3' = -3.97^\circ \tag{13}$$

负号表示折射线的延长线与轴相交于 A 点的左方,它射至光纤的上半部的界面下,如图4所示,$|\alpha_3'| < 21.04^\circ$,所以此光线可以经过全反射在光纤中传播.

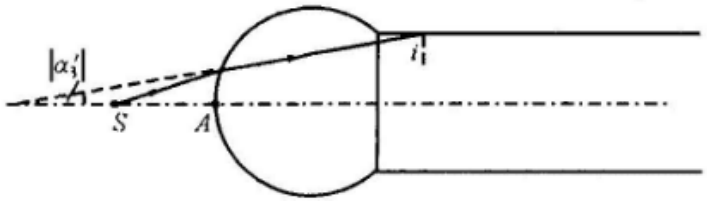


图4